23CSE341 WIRELESS SENSOR NETWORKS

LAB EVALUATION 2 GUIDELINES

Design and Implementation of an Advanced Hybrid Wireless Sensor Network using ESP32 with BLE Localization, Wi-Fi MAC, Dynamic Scheduling, Data Aggregation and Adaptive Routing

1. Objective

Design and implement a **4-node Wireless Sensor Network (WSN)** using ESP32 that integrates:

- BLE-based localization
- Wi-Fi-based MAC protocol
- Broadcast and unicast communication
- Dynamic schedule-based communication
- Data aggregation
- Multi-hop routing
- Route failure/recovery
- Performance measurement

The implementation must demonstrate how different WSN protocols operate together in a single network.

2. Node Configuration

Use a minimum of **4 ESP32 nodes**.

Node	Role	Major Functions
Node 1 (N1)	Sensor Node	Sensing + BLE + Wi-Fi + Data Transmission
Node 2 (N2)	Sensor / Relay Node	Sensing + BLE + Wi-Fi + Data Forwarding
Node 3 (N3)	Cluster Head / Aggregator / Router	Data Aggregation + Routing + Scheduling + Forwarding
Node 4 (N4)	Sink / Gateway	Data Collection + Monitoring + Dashboard

The experiment must use a **hybrid BLE + Wi-Fi architecture**.

BLE must be used for:

- Neighbor discovery
- BLE advertising/scanning
- RSSI measurement
- Approximate distance estimation
- Node proximity/nearest-node identification
- Node localization
- Localization-error calculation

Wi-Fi must be used for:

- MAC/channel access mechanism
- Broadcast communication
- Unicast communication
- Dynamic scheduling
- Sensor-data transmission
- Data aggregation
- Multi-hop routing
- Route recovery
- Sink/gateway communication

4. Requirements

a. **BLE Localization**

Students must:

- Implement BLE advertising and scanning.
- Discover neighbouring nodes using BLE.
- Measure RSSI between nodes.
- Estimate approximate distance from RSSI.
- Determine the approximate location/proximity of a sensor node.
- Calculate localization error.

Students must show the **measured RSSI, estimated distance and localization result** during the demonstration.

b. **Wi-Fi MAC Protocol**

- Implement **channel sensing → random backoff → transmission → ACK**.
- Implement **timeout and retransmission** when ACK is not received.
- Collision/contention handling

Students must show at least one situation where **two nodes attempt transmission at approximately the same time** and explain how their MAC mechanism handles it.

c. **Broadcast & Unicast**

Broadcast must be used for messages intended for multiple nodes, such as:

- Network-control messages
- Schedule information
- Route-control information
- Network status messages

Unicast must be used for:

- Sensor-data transmission
- Next-hop communication
- Aggregated-data transmission
- ACK messages

The destination node and successful packet delivery must be clearly demonstrated.

d. **Dynamic Schedule-Based Communication**
   - Implement **TDMA/time-slot-based transmission**.
   - Nodes transmit only during their assigned slots.
   - Demonstrate unused-slot/sleep operation.
   - The schedule must not remain completely static. Students must explain why and how the schedule was updated.
   - Should demonstrate:
     - Slot assignment
     - Slot-based transmission
     - Unused slot handling
     - Schedule update
     - Sleep/wake operation

e. **Data Aggregation**
   - Node 3 must combine data received from N1 and N2.
   - Forward **aggregated data** instead of forwarding every packet individually.
   - Calculate **data reduction percentage**.
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f. **Adaptive Multi-Hop Routing**

Students must implement multi-hop routing between the sensor nodes and sink.

Route selection may consider parameters such as:

- RSSI/link quality
- Hop count
- Packet loss
- Successful ACK rate
- Link availability

The system must maintain sufficient information about neighboring nodes/routes to select the next hop.

Route Failure and Recovery

Students must demonstrate all of the following:

1. Link failure
2. Failure detection
3. Route update
4. Alternate-route selection
5. Successful packet delivery through the alternate route

g. **Performance Analysis**
Students must measure and analyze **at least four** of the following metrics:

- Packet Delivery Ratio (PDR)
- Packet loss
- End-to-end delay
- Throughput
- Number of retransmissions
- Localization error
- Data reduction percentage

Students must provide:

- Measured values
- Calculations
- Tables/graphs where appropriate
- Brief interpretation of the results

5. Demonstration Flow

Students must demonstrate the following integrated sequence:

BLE Discovery
↓
RSSI Measurement
↓
Localization
↓
Wi-Fi Initialization
↓
Dynamic Schedule Assignment
↓
Channel Sensing
↓
Random Backoff
↓
Scheduled Transmission

↓
Unicast Communication
↓
Data Aggregation
↓
Multi-Hop Routing
↓
Sink
↓
Link Failure
↓
Failure Detection
↓
Alternate Route Selection
↓
Route Recovery
↓
Performance Analysis

Deliverables

- Circuit/network setup
- ESP32 source code
- Architecture diagram
- Flowchart
- Output screenshots
- Performance results
- Short Technical report

The evaluation will emphasize **protocol implementation, integration, experimentation and analysis**, rather than simply showing that the ESP32 can communicate.

Students must develop a simple real-time dashboard to visualize node status, BLE-based localization, sensor data, routing path, aggregated data, MAC activity, and key performance metrics.

EVALUATION RUBRICS

S.No.	Evaluation Component	Criteria for Evaluation	Marks
1	**WSN Architecture & ESP32 Setup**	4 ESP32 nodes configured correctly; sensor, cluster-head/aggregator, router and sink roles clearly implemented; proper network topology and addressing	**2**
2	**BLE-Based Localization**	BLE advertising/scanning, RSSI measurement, distance estimation, nearest-node identification and localization-error calculation	**3**
3	**Wi-Fi MAC Protocol**	Channel sensing, back-off mechanism, ACK, timeout, retransmission and collision-handling implemented and demonstrated	**3**
4	**Dynamic Schedule-Based Communication**	TDMA-style slot allocation, dynamic schedule update, unused-slot handling and sleep/wake operation	**2**
5	**Broadcast & Unicast Communication**	Correct broadcast of control/schedule messages and reliable unicast sensor-data transmission with destination/ACK handling	**1**
6	**Data Aggregation**	Aggregator receives data from multiple nodes, applies aggregation function and demonstrates reduction in transmitted data	**2**
7	**Adaptive Multi-Hop Routing**	Neighbor discovery/table, route metric, multi-hop forwarding, route selection and alternate-route recovery after link failure	**2**
8	**Performance Analysis**	Minimum 4 metrics measured and analyzed: PDR, packet loss, delay, throughput, retransmission, localization error, etc.	**2**
9	**Technical Demonstration&Viva**	Technical explanation and Viva	**3**
	TOTAL		**20**